

MATH 231 – Applied Linear Algebra

A Historical Progression Through the Course Material

Fall 2026 – companion to the syllabus progression

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How to read this document

Linear algebra is almost always taught *backwards* relative to how it was discovered. Students meet the clean axioms first (a vector space, a linear map) and only later the messy computations (elimination, determinants, eigenvalues). History ran the other way: people computed with systems of equations for two thousand years, manipulated determinants for a century and a half, and multiplied matrices for decades *before* anyone wrote down the axioms of a vector space. The abstraction was the last thing to arrive, not the first.

This companion therefore has two jobs. It follows the **same four arcs and twelve units** as `MATH231_progression_2026-07-01.tex`, so each section can be read alongside the unit it historicizes. But it also lets you see the true chronological plot underneath the pedagogical one: determinants before matrices; least squares (Unit 10) driving elimination (Unit 7) and the very idea of a condition number (Unit 6); celestial mechanics and psychometrics driving eigenvalues and the SVD (Unit 11). Marginal tags such as [\[Units 7, 10\]](#) point back to where the topic lives in the progression.

A recurring theme is worth stating up front: **almost every tool in this course was invented to solve a physical or data-analytic problem, not for its own sake**. Elimination came from taxation and surveying; least squares from a lost asteroid; eigenvalues from the wobble of planets and spinning tops; the SVD twice over from geometry and then from psychology; gradient descent from orbit-fitting; PageRank from ranking the early Web. The abstract structure is the residue left behind after the applications were understood.

Arc I – Foundations

1. Vectors and Vector Spaces [\[Unit 1\]](#)

1.1 The object arrived long before the axioms

The modern definition of a vector space — a set closed under addition and scalar multiplication satisfying a short list of axioms — is the *end* of a long story, not its beginning. Its true originator is **Hermann Grassmann (1844)**, a Prussian schoolteacher who published *Die lineale Ausdehnungslehre* (“The Theory of Linear Extension”). Grassmann, trying to give algebraic meaning to geometric magnitudes of arbitrary dimension, introduced essentially everything in Unit 1: linear combinations, linear independence, subspaces, and *dimension* as an intrinsic invariant — decades before anyone else thought in $\mathbb{R}^n$ for $n > 3$. His book was so far ahead of its idiom that it went almost

entirely unread; a rewritten 1862 edition fared little better in his lifetime ([MacTutor: Grassmann](#); [MacTutor: Abstract linear spaces](#)).

The crisp *axioms* students memorize are due to **Giuseppe Peano (1888)** in *Calcolo geometrico secondo l'Ausdehnungslehre di H. Grassmann*. In a book ostensibly popularizing Grassmann, Peano wrote down, in a single chapter, the axioms of a real vector space in a form a student would recognize today — and then, remarkably, discussed infinite-dimensional examples ([Wikipedia: Vector space – History](#); [MacTutor: Peano](#)). It took another generation — and the needs of functional analysis (**Banach, Wiener, Weyl** in the 1920s) — for the axiomatic viewpoint to become standard.

1.2 Where the word “vector” comes from

The term “vector” (and its companion “scalar”) is not Grassmann’s; it is **William Rowan Hamilton’s (1840s)**, coined while developing quaternions (Unit 3), from the Latin *vehere*, “to carry.” The now-ubiquitous dot and cross products, and the very phrase “vector analysis,” were shaped in the 1880s by **J. Willard Gibbs** and **Oliver Heaviside**, who stripped Hamilton’s quaternions down to the 3-vector calculus physicists actually needed ([Wikipedia: Vector – History](#)). So the course’s opening object carries three distinct lineages: Grassmann’s abstract structure, Hamilton’s name, and Gibbs–Heaviside’s working notation.

2. Inner Products, Norms & Similarity [Unit 2]

2.1 The Cauchy–Schwarz inequality: three names, three settings

The inequality $|\langle x, y \rangle| \leq \|x\| \|y\|$ is really the *Cauchy–Bunyakovsky–Schwarz* inequality, and its triple name records three independent settings. **Augustin-Louis Cauchy** proved the finite-sum version in his 1821 *Cours d’analyse*. **Viktor Bunyakovsky (1859)** extended it to integrals. **Hermann Amandus Schwarz (1888)** rediscovered the integral form in his work on minimal surfaces and gave the proof now taught ([Wikipedia: Cauchy–Schwarz – History](#); [MacTutor: Cauchy](#)). The inequality is exactly what makes “the cosine of the angle between two vectors” well defined — the geometric fact underlying every cosine-similarity computation in the course’s ML demos.

2.2 Norms and the ℓ_p family

The p -norms the unit tabulates ($\ell_1, \ell_2, \ell_\infty$) grew out of two nineteenth-century inequalities: **Otto Hölder’s (1889)** and **Hermann Minkowski’s (1896)**, the latter from *Geometrie der Zahlen*, where Minkowski was studying convex bodies and the “geometry of numbers” ([MacTutor: Minkowski](#)). The abstract notion of a *norm* — a length function axiomatizing all of these at once — and the term “normed space” belong to **Stefan Banach’s (1922)** thesis (with parallel work by Hahn, Helly, and Wiener), the founding document of functional analysis ([Wikipedia: Normed vector space](#)). Thus the taxicab/Euclidean/Chebyshev distinction the unit draws is a compression of a story that runs from number theory to the birth of modern analysis.

2.3 Gram–Schmidt: orthogonalization with a prehistory

The orthogonalization procedure is named for **Jørgen Pedersen Gram (1883)** and **Erhard Schmidt (1907)**, but neither invented it: essentially the same construction appears in work of **Laplace** and **Cauchy** decades earlier, and Schmidt himself credited Gram ([Wikipedia: Gram–Schmidt](#)). Gram met it while studying least-squares approximation by orthogonal functions — already tying Unit 2 forward to Unit 10. The distinction between “classical” and “modified” Gram–Schmidt that the

progression defers to Unit 10 is a *twentieth-century, numerical* discovery: the two are identical in exact arithmetic and wildly different in floating point (Unit 6).

3. Complex Numbers & Quaternions [Unit 3]

3.1 Complex numbers were forced, not invented

Nobody set out to invent $\sqrt{-1}$; it was forced on algebraists by the cubic. **Gerolamo Cardano**, in *Ars Magna* (1545), was pushed into writing $5 \pm \sqrt{-15}$ while solving equations and called such quantities “as subtle as they are useless.” **Rafael Bombelli** (1572) gave consistent arithmetic rules for them, noticing they were unavoidable even when the *final* answers were real (the *casus irreducibilis*) (MacTutor: cubic/quartic equations; Wikipedia: Complex number – History).

The *geometric* picture the unit relies on — the complex plane, and multiplication as rotation-and-scaling — came much later and, tellingly, from three people at once: **Caspar Wessel** (1799), **Jean-Robert Argand** (1806), and **Carl Friedrich Gauss**, who coined the term “complex number” and did most to legitimize the plane picture in (1831). **Hamilton** (1837) then removed the last whiff of mysticism by *defining* a complex number as an ordered pair (a, b) of reals with prescribed multiplication — a purely algebraic construction. **Euler’s** formula $e^{i\theta} = \cos \theta + i \sin \theta$ (1748) (anticipated by **Roger Cotes**, 1714) and **de Moivre’s** theorem (1707–1722) supplied the bridge between the algebra and the trigonometry the unit exploits (MacTutor: Euler). The **Fundamental Theorem of Algebra**, that this number system is finally “complete” for polynomials, was first proved (with gaps by modern standards) by Gauss in his 1799 doctoral dissertation (MacTutor: FTA).

3.2 Quaternions: Hamilton’s search for 3-D multiplication

The quaternion enrichment topic has one of the most famous origin stories in mathematics. **Hamilton** spent years trying to build a consistent multiplication on *triples* to do for 3-D space what complex numbers do for the plane — and it is impossible. On 16 October 1843, walking to Broom (Brougham) Bridge in Dublin, he realized he needed *four* dimensions and a *non-commutative* product ($ij = -ji$), and carved $i^2 = j^2 = k^2 = ijk = -1$ into the bridge (MacTutor: Hamilton; Wikipedia: Quaternion – History). The motivating problem — rotations in 3-D — is exactly why quaternions survive today in computer graphics and robotics, where they avoid the “gimbal lock” of Euler angles. Non-commutativity, first met here, is the student’s first warning that matrix multiplication (Unit 4) need not commute.

Arc II – Matrices & Continuous Math

4. Matrices: Algebra, Determinant & Trace [Unit 4]

4.1 Determinants came first — by 150 years

One of the great inversions in the history of the subject: the *determinant* predates the *matrix* by about a century and a half. Working independently, **Seki Takakazu** in Japan (1683) and **Gottfried Wilhelm Leibniz** (1693, in a letter to l’Hôpital) developed determinant-like expressions to decide when systems of linear equations have solutions (MacTutor: Matrices and determinants). **Gabriel Cramer** (1750) gave the explicit rule now bearing his name while studying which plane

curve passes through given points. **Pierre-Simon Laplace** (1772) gave the cofactor (“Laplace”) expansion. The subject was finally systematized by **Cauchy** (1812), who fixed the modern meaning of the word “determinant,” proved the product theorem $\det(AB) = \det(A)\det(B)$ the unit states, and connected determinants to the eigenvalue work of Unit 11 ([Wikipedia: Determinant – History](#)).

4.2 The matrix as an object: Sylvester and Cayley

The word “matrix” — Latin for “womb,” the thing from which determinants are born — was coined by **James Joseph Sylvester** (1850). But the decisive step, treating a matrix as a *single algebraic object* with its own addition, multiplication, and inverse, is **Arthur Cayley**’s *A Memoir on the Theory of Matrices* (1858). There Cayley defined matrix multiplication so that it represents *composition of linear substitutions* (the Unit 8 viewpoint, arrived at first!), noted that it is non-commutative, and stated what we call the Cayley–Hamilton theorem ([Wikipedia: Cayley](#); [MacTutor: Cayley](#); and Cayley’s paper via [JSTOR / Phil. Trans. Royal Soc. 148](#) (1858)). It is historically accurate to tell students that *matrix multiplication is defined the way it is because it must mirror the composition of linear maps* — that is the reason Cayley chose the rule.

The word “trace” is a translation of the German *Spur*; the matrix exponential e^A the unit previews belongs to the theory of linear differential equations and, later, to **Sophus Lie**’s theory of continuous groups.

5. Multivariable Calculus: Gradients, Hessians & Taylor [Unit 5]

This unit imports the analytic machinery that Arc IV will need. Its pieces have sharply datable origins. **Brook Taylor** published his expansion in *Methodus Incrementorum Directa et Inversa* (1715), though special cases were known to Gregory and Newton ([MacTutor: Taylor](#)). *Partial* derivatives and the equality of mixed partials (Clairaut’s theorem) come from the 18th-century analysts **Euler**, **Alexis Clairaut**, and **d’Alembert**, wrestling with the vibrating-string and heat problems.

The two named objects of the unit are eponymous. The **Hessian** matrix of second partials is named for **Ludwig Otto Hesse** (mid-1800s), who used it studying curves and surfaces ([Wikipedia: Hessian](#)); the closely related **Jacobian** is **Carl Gustav Jacob Jacobi**’s (1841) ([MacTutor: Jacobi](#)). The symbol ∇ (“nabla”/“del”) for the gradient was introduced by **Hamilton** and named “gradient” in the vector-calculus tradition of Maxwell and Gibbs. Note the forward reference the progression itself flags: the gradient and Hessian are *defined* here, but the meaning of Hessian *definiteness* must wait for the spectral theory of Unit 11 — historically, too, definiteness of quadratic forms (Gauss, Sylvester) matured alongside eigenvalue theory, not calculus.

6. Numerical Foundations: Conditioning & Stability [Unit 6]

This is the most modern unit in the whole course: its central concepts are essentially products of the *electronic-computer era*, invented because people were, for the first time, running thousands of arithmetic operations and watching errors accumulate.

The **condition number** — how much a problem amplifies input error, independent of any algorithm — was introduced by **Alan Turing** (1948) in “Rounding-off errors in matrix processes,” and in parallel by **John von Neumann** and **Herman Goldstine** (1947) in their analysis of inverting large matrices for early computers ([MacTutor: Turing](#); [Wikipedia: Condition number](#)). Turing’s

same paper is one of the birthplaces of the LU viewpoint of Unit 7. The distinction between a well-conditioned *problem* and a stable *algorithm* — and the technique of *backward error analysis* that makes it precise — is the life’s work of **James H. Wilkinson** in the 1950s–60s, crystallized in *Rounding Errors in Algebraic Processes* (1963) and *The Algebraic Eigenvalue Problem* (1965) (Wikipedia: Wilkinson). The floating-point system itself was standardized only in (1985) as IEEE 754, driven largely by **William Kahan** (Wikipedia: IEEE 754). On the complexity side, “big- O ” notation is older and comes from analytic number theory — **Paul Bachmann** (1894) and **Edmund Landau** (1909) — and was imported into the analysis of algorithms by **Donald Knuth** in the 1960s (Wikipedia: Big-O notation).

Arc III – Core Matrix Theory

7. Linear Systems & Elimination [Unit 7]

7.1 The oldest algorithm in the course

Gaussian elimination is by far the oldest technique in MATH 231 — roughly two millennia older than the man it is named for. Chapter 8 (*Fangcheng*, “rectangular arrays”) of the Chinese *Nine Chapters on the Mathematical Art* (*Jiǔzhāng Suànrshù*), compiled during the Han dynasty around the (1st century BCE), solves systems of up to five equations by exactly the row-reduction we teach, laying coefficients out as a rectangular array of counting rods and eliminating column by column — and freely using negative numbers to do it (Wikipedia: Gaussian elimination – History; MacTutor: The Nine Chapters).

Gauss himself came to elimination not as pure algebra but through *astronomy and geodesy*: he needed to solve the overdetermined normal equations of least squares (Unit 10) to fit orbits and survey the kingdom of Hanover, and described a systematic elimination in *Theoria Motus* (1809) and later geodetic work. The name “Gaussian elimination” is a 20th-century retronym. The “Gauss–Jordan” refinement is due to the geodesist **Wilhelm Jordan** (1888) (not the mathematician Camille Jordan), again motivated by surveying computations.

7.2 LU factorization: elimination as a product

Recognizing that elimination *is* a factorization $A = LU$ into lower- and upper-triangular factors — and analyzing when pivoting is needed for stability (the tie back to Unit 6) — is again modern, belonging to **Turing** (1948) and to the mid-century numerical-analysis school (Doolittle, Crout, and the Polish astronomer **Banachiewicz** contributed variants) (Wikipedia: LU decomposition). The historical order is worth stressing to students: the *algorithm* is ancient, but the *factorization interpretation* that makes it composable and analyzable is younger than the computer.

8. Linear Transformations [Unit 8]

The idea that a matrix *is* the coordinate shadow of a coordinate-free *linear map* $T : V \rightarrow W$ was, historically, present at the creation of matrix algebra: as noted in Unit 4, **Cayley** (1858) defined matrix multiplication precisely to encode the composition of “linear substitutions.” The coordinate-free *map* viewpoint — kernels, images, and the behavior of maps independent of a chosen basis — matured with **Grassmann** and **Peano** (Unit 1) and became the organizing principle

of the subject in the 20th century, through **Emmy Noether**’s structural algebra and the textbooks it inspired ([Wikipedia: Linear map](#)). The word “rank” (*Rang*) for the dimension of the image is **Ferdinand Georg Frobenius**’s (1878) ([MacTutor: Frobenius](#)), and the change-of-basis/similarity relation $B = P^{-1}AP$ is what Cauchy and Jordan were implicitly using in their eigenvalue work. The **Rank–Nullity theorem** is a direct descendant of Grassmann’s dimension formula for subspaces.

9. The Four Fundamental Subspaces [Unit 9]

The individual facts here — column space, null space, row space, left null space, and the orthogonality between the pairs — are consequences of nineteenth-century rank theory (Frobenius, Sylvester). But the *packaging* as “the four fundamental subspaces” and “the Fundamental Theorem of Linear Algebra” is a modern, explicitly *pedagogical* creation of **Gilbert Strang**, argued most directly in his 1993 *American Mathematical Monthly* article “The Fundamental Theorem of Linear Algebra” ([JSTOR: Strang, Amer. Math. Monthly 100 \(1993\)](#); [Wikipedia: Fundamental theorem of linear algebra](#)). This is a good place to tell students that not every “theorem” is a discovery of a new fact — sometimes it is the discovery of the *right way to see* old facts, and the payoff is that projection and least squares (Unit 10) and the geometry of the SVD (Unit 11) fall out almost for free.

Arc IV – Advanced Methods & Applications

10. Orthogonality, Projections, Least Squares & Factorizations [Unit 10]

10.1 Least squares: a lost asteroid and a priority dispute

Least squares is the historical engine that drives much of this course, and it was born from a concrete emergency in astronomy. On 1 January 1801 the astronomer Piazzi observed the dwarf planet *Ceres* for 41 days, then lost it in the Sun’s glare. **Gauss**, age 24, recovered it by fitting an orbit to the sparse, noisy data — and the method he used was least squares. He published it in *Theoria Motus* (1809), claiming to have used it since 1795, but **Adrien-Marie Legendre** had *published* it first, in an 1805 appendix under the name *méthode des moindres carrés*. The resulting priority dispute is one of the famous quarrels in the history of mathematics ([Wikipedia: Least squares – History](#); [MacTutor: Legendre](#)). Gauss went further, connecting least squares to the normal (Gaussian) distribution and thereby founding the statistical theory of estimation — which is why linear regression, the Unit 10 core application, *is* least squares.

10.2 The factorizations that make it stable

The unit’s factorizations are 20th-century numerical refinements of Gauss’s idea, each solving a stability problem the normal equations $A^T Ax = A^T b$ create (the $\kappa(A^T A) = \kappa(A)^2$ blow-up of Unit 6):

- **Cholesky factorization** $A^T A = LL^T$ is due to **André-Louis Cholesky** around (1910), a French military geodesist solving normal equations for triangulation surveys. He was killed in 1918; his method was published posthumously in 1924 ([Wikipedia: Cholesky](#); [MacTutor: Cholesky](#)). A pure application of least-squares surveying, exactly the setting of the unit.

- **Householder QR** uses the reflections introduced by **Alston Householder (1958)** in “Unitary triangularization of a nonsymmetric matrix” — expressly to compute more stably than Gram–Schmidt ([Wikipedia: Householder transformation](#); [MacTutor: Householder](#)). The alternative, **Givens** rotations, appeared the same year. **Modified Gram–Schmidt**, as noted in Unit 2, is the same 1907 procedure re-analyzed for floating point.

The **Discrete Cosine Transform** the unit uses to contrast a *fixed* orthonormal basis with the data-adaptive SVD basis was introduced by **Nasir Ahmed**, T. Natarajan, and K.R. Rao (**1974**), and became the mathematical heart of the JPEG image standard (**1992**) ([Wikipedia: DCT – History](#)).

11. Eigenvalues, Eigenvectors & the SVD [Unit 11]

11.1 Eigenvalues: wobbling planets and spinning tops

Eigenvalues entered mathematics through *physics*, long before matrices existed to house them. In the 18th century **Euler** studied the principal axes of rotation of a rigid body (the directions in which a spinning top rotates cleanly), and **Lagrange** and **Laplace** studied the *secular equation* — “secular” from *saeculum*, a century — governing the slow, long-term perturbations of the planetary orbits. The condition for a stable orbit is exactly an eigenvalue condition ([Wikipedia: Eigenvalues – History](#)). **Cauchy (1829)** unified this: studying quadratic forms, he proved that a (real) symmetric matrix has real eigenvalues and orthogonal principal axes — the **spectral theorem** in embryo, and the source of the Hessian-definiteness analysis promised back in Unit 5. The name is a translation curiosity: the German *Eigenwert* (“own/characteristic value”) was fixed by **David Hilbert (1904)** in his study of integral operators, where he also introduced “spectrum” (*Spektrum*); English kept the German prefix, giving the hybrid “eigenvalue.” That eigenvalues of large matrices must be found *iteratively* is not a computational accident but a theorem: the **Abel–Ruffini theorem (Paolo Ruffini (1799), Niels Henrik Abel (1824), completed by Galois)** says the degree- ≥ 5 characteristic polynomial has no general solution in radicals ([Wikipedia: Abel–Ruffini](#)). The modern **QR algorithm** that actually does the iteration was found independently by **John G.F. Francis (1961)** and **Vera Kublanovskaya (1961)** ([Wikipedia: QR algorithm](#)).

11.2 The SVD: discovered (at least) three times

The Singular Value Decomposition has one of the richest multiple-discovery histories in mathematics. **Eugenio Beltrami (1873)** and **Camille Jordan (1874)** found it independently while studying bilinear forms; **James Joseph Sylvester (1889)** rediscovered it again; **Erhard Schmidt (1907)** produced the infinite-dimensional (integral-operator) version and, crucially, the idea of *best low-rank approximation* ([Wikipedia: SVD – History](#); G.W. Stewart’s survey “On the Early History of the Singular Value Decomposition,” *SIAM Review* 35 (1993)). The result the unit calls **Eckart–Young** — that truncating the SVD gives the optimal low-rank approximation — was proved by **Carl Eckart** and **Gale Young (1936)** in a *psychology journal*, *Psychometrika*, because the motivating problem was factor analysis of psychological test data (**Eckart & Young, Psychometrika** 1 (1936)). The practical algorithm for computing the SVD is **Gene Golub** and **William Kahan’s (1965)**.

11.3 PCA, PageRank, and spectral graphs: eigenvalues escape physics

The applications in this unit show the same eigen-machinery migrating into statistics and data:

- **Principal Component Analysis** was introduced geometrically by **Karl Pearson (1901)** (“On lines and planes of closest fit to systems of points in space”) and independently, with the modern name and statistical framing, by **Harold Hotelling (1933)** — again from *psychometrics* ([Wikipedia: PCA – History](#); [Pearson 1901, Phil. Mag.](#)). The “Eigencats/Eigenfaces” project descends directly from this line (Turk & Pentland, 1991).
- The **Perron–Frobenius theorem** — a nonnegative matrix has a positive dominant eigenvalue with a nonnegative eigenvector — is **Oskar Perron’s (1907)** and **Georg Frobenius’s (1912)** ([Wikipedia: Perron–Frobenius](#)). It is the mathematical guarantee that **PageRank** works: **Larry Page** and **Sergey Brin (1998)** defined a web page’s importance as its coordinate in the dominant eigenvector of a stochastic “random surfer” matrix (Brin & Page, “The Anatomy of a Large-Scale Hypertextual Web Search Engine,” WWW7 1998; Page, Brin, Motwani & Winograd, “The PageRank Citation Ranking,” 1999).
- **Spectral graph theory** begins with **Gustav Kirchhoff’s (1847)** matrix-tree theorem for electrical networks (the origin of the graph Laplacian $L = D - W$), and the clustering use rests on **Miroslav Fiedler’s (1973)** “algebraic connectivity” and the eigenvector now called the Fiedler vector ([Wikipedia: Algebraic connectivity](#)), later turned into the normalized-cut clustering of Shi & Malik (2000).
- **Latent Semantic Analysis** (Demo #3) applies the truncated SVD to a TF-IDF term–document matrix: the SVD/LSA method is Deerwester, Dumais, Landauer, Furnas & Harshman (1990), and the inverse-document-frequency weighting is **Karen Spärck Jones’s (1972)** ([Wikipedia: LSA – History](#)).

12. Numerical Optimization & the ML Capstone [Unit 12]

12.1 Descent methods, from Cauchy to the present

The capstone’s optimization methods again reach back to concrete nineteenth-century problems. **Gradient descent** (“steepest descent”) is **Cauchy’s (1847)**, proposed in a short note precisely for solving the systems of equations arising in *astronomical orbit computation* — the same problem family that produced least squares ([Wikipedia: Gradient descent – History](#)). **Newton’s method** descends from **Isaac Newton (c. 1669)** and **Joseph Raphson (1690)**; its use inside optimization makes each step a linear $\text{solve } H\Delta = -\nabla f$, tying the capstone straight back to Unit 7.

For nonlinear least squares (the trilateration project), **Gauss–Newton** descends from Gauss’s own orbit work; the damped variant is **Kenneth Levenberg’s (1944)** and **Donald Marquardt’s (1963)**, the latter developed at DuPont for fitting chemical models to data ([Wikipedia: Levenberg–Marquardt](#)). The quasi-Newton **BFGS** method is the 1970 work of **Broyden, Fletcher, Goldfarb** and **Shanno (Wikipedia: BFGS)**. The convergence-depends-on-conditioning theme that closes the course is exactly the Turing/Wilkinson insight of Unit 6 applied to iteration.

12.2 The machine-learning payoff

The learning algorithms the course ends on are recent but rest entirely on the linear-algebra spine built in Arcs I–III:

- The **Support Vector Machine** grows out of the statistical learning theory of **Vladimir Vapnik** and **Alexey Chervonenkis** in the 1960s–70s; the kernelized, maximum-margin form the unit

teaches is Boser, Guyon & Vapnik (1992) and the soft-margin version is Cortes & Vapnik (1995) (Wikipedia: SVM – History).

- *k*-means (Project #1) traces to **Hugo Steinhaus (1956)**, **Stuart Lloyd** (Bell Labs, (1957), published 1982), and **James MacQueen (1967)**, who coined the name; *k*-nearest-neighbors to **Fix & Hodges (1951)** and **Cover & Hart (1967)** (Wikipedia: *k*-means; Wikipedia: *k*-NN).
- The **static word-embedding** geometry (Demo #9) is the most recent idea in the course — *word2vec* (Mikolov et al., (2013)) and *GloVe* (Pennington, Socher & Manning, (2014)) — yet the analogy arithmetic king – man + woman $\approx$ queen is *pure* Unit 2 linear algebra: vector addition and cosine similarity (Mikolov et al., “Efficient Estimation of Word Representations,” arXiv:1301.3781; Pennington et al., GloVe, EMNLP 2014). It is a fitting close: the newest tools in machine learning cash out, geometrically, as the oldest ideas in the course.

A one-page chronology

Date	Milestone	Course unit
c. 1st c. BCE	<i>Nine Chapters</i> , Ch. 8: elimination on rod arrays	7
1545	Cardano meets $\sqrt{-1}$ in the cubic (<i>Ars Magna</i>)	3
1683/1693	Seki; Leibniz: determinants	4
1715	Taylor's theorem	5
1748	Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$	3
1799/1824	Ruffini–Abel: quintic unsolvable in radicals	11
1801/1805/1809	Ceres; Legendre & Gauss: least squares	10
1829	Cauchy: real eigenvalues of symmetric forms (spectral theorem)	11
1843	Hamilton: quaternions	3
1844/1888	Grassmann; Peano: vector spaces & axioms	1
1847	Cauchy: steepest descent / Kirchhoff: graph Laplacian	12 / 11
1850/1858	Sylvester names “matrix”; Cayley's matrix algebra	4, 8
1873/1874	Beltrami & Jordan: the SVD	11
1901/1933	Pearson; Hotelling: PCA	11
1907/1912	Perron–Frobenius theorem	11
1907	Gram–Schmidt (Schmidt); low-rank approximation	2, 11
1910/1924	Cholesky factorization (geodesy)	10
1936	Eckart–Young low-rank optimality (<i>Psychometrika</i>)	11
1947/1948	von Neumann–Goldstine; Turing: conditioning, LU	6, 7
1958	Householder reflections	10
1961	Francis & Kublanovskaya: QR eigenvalue algorithm	11
1963/1965	Wilkinson: backward error analysis	6
1965	Golub–Kahan: computing the SVD	11
1972/1974	Spärck Jones: IDF / Ahmed et al.: DCT	11, 10
1990/1995	LSA / soft-margin SVM	11, 12
1998	Page & Brin: PageRank	11
2013/2014	word2vec; GloVe embeddings	11

Sources & further reading (all links)

History surveys & general references.

- MacTutor, *Matrices and determinants*: https://mathshistory.st-andrews.ac.uk/HistTopics/Matrices_and_determinants/
- MacTutor, *Abstract linear spaces*: https://mathshistory.st-andrews.ac.uk/HistTopics/Abstract_linear_spaces/
- MacTutor, *The Nine Chapters*: https://mathshistory.st-andrews.ac.uk/HistTopics/Nine_chapters/
- MacTutor, *Fundamental theorem of algebra*: https://mathshistory.st-andrews.ac.uk/HistTopics/Fund_theorem_of_algebra/
- MacTutor, *Quadratic, cubic and quartic equations*: https://mathshistory.st-andrews.ac.uk/HistTopics/Quadratic_etc_equations/

- G.W. Stewart, “On the Early History of the Singular Value Decomposition,” *SIAM Review* 35 (1993): <https://www.jstor.org/stable/2132388>

Biographies (MacTutor).

- Grassmann: <https://mathshistory.st-andrews.ac.uk/Biographies/Grassmann/> Peano: <https://mathshistory.st-andrews.ac.uk/Biographies/Peano/>
- Cauchy: <https://mathshistory.st-andrews.ac.uk/Biographies/Cauchy/> Hamilton: <https://mathshistory.st-andrews.ac.uk/Biographies/Hamilton/>
- Cayley: <https://mathshistory.st-andrews.ac.uk/Biographies/Cayley/> Euler: <https://mathshistory.st-andrews.ac.uk/Biographies/Euler/>
- Legendre: <https://mathshistory.st-andrews.ac.uk/Biographies/Legendre/> Jacobi: <https://mathshistory.st-andrews.ac.uk/Biographies/Jacobi/>
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- Determinant: <https://en.wikipedia.org/wiki/Determinant#History> Complex number: https://en.wikipedia.org/wiki/Complex_number#History
- Quaternion: <https://en.wikipedia.org/wiki/Quaternion#History> Cauchy–Schwarz: https://en.wikipedia.org/wiki/Cauchy%E2%80%93Schwarz_inequality#History
- Gram–Schmidt: https://en.wikipedia.org/wiki/Gram%E2%80%93Schmidt_process#History Normed space: https://en.wikipedia.org/wiki/Normed_vector_space#History
- Gaussian elimination: https://en.wikipedia.org/wiki/Gaussian_elimination#History LU decomposition: https://en.wikipedia.org/wiki/LU_decomposition#History
- Condition number: https://en.wikipedia.org/wiki/Condition_number IEEE 754: https://en.wikipedia.org/wiki/IEEE_754#History
- Big-O notation: https://en.wikipedia.org/wiki/Big_O_notation#History Linear map: https://en.wikipedia.org/wiki/Linear_map#History

- Eigenvalues: https://en.wikipedia.org/wiki/Eigenvalues_and_eigenvectors#History Abel–Ruffini: https://en.wikipedia.org/wiki/Abel%E2%80%93Ruffini_theorem
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- PCA: https://en.wikipedia.org/wiki/Principal_component_analysis#History Perron–Frobenius: https://en.wikipedia.org/wiki/Perron%E2%80%93Frobenius_theorem
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- Hessian: https://en.wikipedia.org/wiki/Hessian_matrix Fundamental theorem of linear algebra: https://en.wikipedia.org/wiki/Fundamental_theorem_of_linear_algebra

Note on reliability. Dates for multiply-discovered results (the SVD, least squares, Gram–Schmidt) are inherently contested; where a priority dispute exists it is flagged in the text. MacTutor (St Andrews) and the cited primary papers are the authoritative sources; Wikipedia links are included for convenient orientation and their own reference lists, not as the final word.